I.MA.TEC.
HONEYCOMB & DERIVATI

Typical values for aluminium honeycomb Alloys 5052 – 5056 - 3003										
Type				Bare Compression		Plate Shear		Plate Shear		Crush PSI
Alloy	Cell inch	Sheet inch	Weight PCF	Strength PSI	Mod. KSI	Stength PSI“L”direc	Strength PSI“W”direc	Mod. KSI“L”direc	Mod. KSI"W"direc	
5052	1/8	.0007	3.1	270	75	210	130	45	22	130
5052	1/8	.001	4.5	520	150	340	220	70	31	260
5052	1/8	.0015	6.1	870	240	505	320	98	41	450
5052	1/8	.002	8.1	1400	350	725	455	135	54	750
5052	1/8	.003	10.0	2000	-	960	530	175	65	820
5052	3/16	.0007	2.0	150	34	110	60	80	46	60
5052	3/16	.001	3.1	270	75	210	125	45	22	130
5052	3/16	.0015	4.4	510	145	330	215	68	30	250
5052	1/4	.001	2.3	175	45	140	85	32	16	75
5052	1/4	.002	4.2	490	140	320	200	66	29	230
5052	1/4	.0025	5.2	680	190	410	265	82	35	335
5056	1/8	.001	4.5	620	185	440	255	70	28	320
5056	3/16	.001	3.1	380	97	265	150	45	20	170
5056	3/16	.0015	4.4	620	180	425	245	68	27	310
3003	1/4	.003	5.2	610	148	345	215	63	31	245
3003	1/4	.002	4.2	440	110	305	178	57	28	165
3003	3/8	.003	3.6	325	92	200	130	40	20	120
3003	3/8	.002	2.7	220	58	165	110	35	17	80
3003	1/2	.003	2.4	165	40	125	70	25	15	60
3003	2/3	.003	2.1	125	28	95	60	18	10	50
3003	3/4	.003	1.8	110	24	85	55	16	8	45

The above data are indicative

Denomination of types of aluminium alloy honeycomb

Alloy – cell – sheet - weight
Alloy: type of aluminium alloy used
Cell: cell size in inches – e.g. 1/8 inch = (1/8 x 25,4) = 3.175 mm
Sheet: Sheet thickness in inches – e.g. .001 inch = (.001 x 25.4) = 0.0254mm
Weight: density in pounds per cubic foot – e.g. 3.1 pcf = (3.1 x 16) = 49.6 kg / m^3